

Differentiation Rules: Power, Product, Quotient, Chain



Power, sum, and constant rules

1 Differentiate each function:

a $f(x) = 5x^4 - 3x^2 + 7$

b $f(x) = \frac{1}{x^3}$

c $f(x) = 4\sqrt{x}$

Product and quotient rules

2 Differentiate $f(x) = (2x + 1)(x^2 - 3)$ using the product rule.

3 Differentiate $f(x) = \frac{x^2 + 1}{x - 2}$ using the quotient rule.

4 Differentiate $f(x) = x^2 \sin x$.

The chain rule

5 Differentiate each function using the chain rule:

a $f(x) = (3x^2 - 5)^4$

b $f(x) = \sin(2x^3)$

c $f(x) = e^{-x^2}$

d $f(x) = \ln(x^2 + 1)$

6 Differentiate $f(x) = \sqrt{4x + \cos x}$.

Combined rules

7 Differentiate $f(x) = \frac{\sin x}{x^2}$ using the quotient rule.

8 Differentiate $f(x) = e^{2x} \cos(3x)$ using the product and chain rules together.

9 Differentiate $f(x) = \left(\frac{x+1}{x-1}\right)^3$ using the chain rule together with the quotient rule.

Tangent lines using derivatives

10 Find the equation of the tangent line to $f(x) = x^2 - 3x + 1$ at $x = 2$.

11 Find the point(s) on the curve $y = x^3 - 3x$ where the tangent line is horizontal.

Higher-order derivatives

12 Find $f''(x)$ for $f(x) = x^4 - 6x^2 + 2x$.

13 Find $f''(x)$ for $f(x) = \sin(2x)$.

Normal lines

14 Find the equation of the line normal (perpendicular) to $f(x) = x^2$ at $x = 2$.